Activity-2 Scrum Report

EcoDiscovery – Interactive Ecosystem Learning Platform

ENSE-281: Software Engineering Management

Group G:
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Project Sponsor: Dr. Tim Maciag

Problem Definition

- Traditional learning methods often struggle to effectively teach complex ecological concepts to young students. Static diagrams and text-based materials do not always provide engaging ways for students to understand how ecosystems function.
- There is a need for interactive educational tools that allow students to explore ecosystems visually and learn about environmental relationships and conservation in a more engaging and intuitive way.

Project Definition

- EcoDiscovery is an interactive educational web application designed to help elementary school students explore freshwater ecosystems through discovery-based learning.
- The system allows students to interact with a visual ecosystem environment, identify animals, and learn ecological facts through engaging gameplay mechanics.
- This approach aims to improve understanding of ecosystems while encouraging curiosity and environmental awareness.

Key System Requirements

- **Functional Requirements**
- • Users can explore an ecosystem environment
  • Users can click animals to discover educational information
  • The system unlocks stickers when animals are correctly identified
  • The system tracks discovery progress and achievements
- **Non-Functional Requirements**
- • Simple and intuitive interface suitable for Grade 3 students
  • Visually engaging educational experience
  • Responsive interaction with minimal delay

Project Scope

In Scope

- • Interactive freshwater ecosystem exploration
  • Animal discovery and educational information display
  • Sticker book system for tracking discoveries
  • Progress tracking and visual feedback
  • Environmental awareness interaction (garbage cleanup feature)

Out of Scope

- • Multiplayer functionality
  • Complex ecological simulation systems
  • Integration with external educational platforms

Stakeholder Engagement

STAKEHOLDER REGISTER				
Project Name	Ecosystem Discovery Application			
Name	**Project Role**	**Level of Power**	**Level of Interest**	**Level of Support**
Elementary School Children (Ages 7–9)	Primary End Users of the Application	Low	High	Supportive
Elementary School Teachers	Secondary Users (classroom facilities)	Medium	High	Supportive
Parents & Guardian	Secondary Users (home learning support)	Low-Medium	Medium	Supportive
Class Instructor (Tim Maciag)	Academic evaluator and project sponsor	High (Internal Control)	High	Supportive
Development Team (Jeremiah, Rayansh, Aubin, Abriana)	Designers and developers of the system	Medium	Medium	Supportive
School Administration	Potential Institutional adopter	Low (current scope	Low	Neutral

System Architecture (MVC)

The system follows the **Model-View-Controller (MVC)** design pattern.

Model

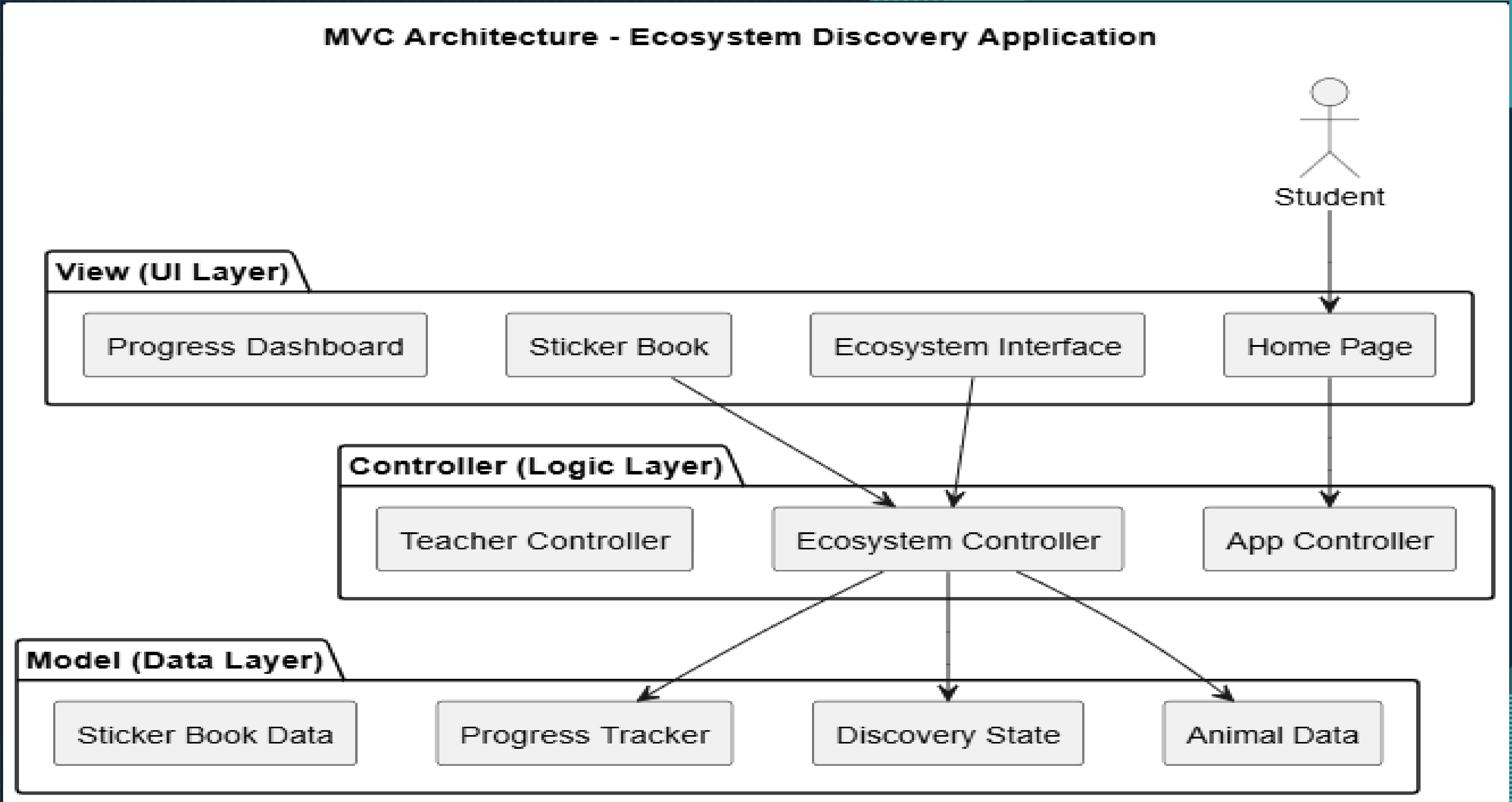
- Stores animal data and discovery progress

View

- Displays the ecosystem interface and user interactions

Controller

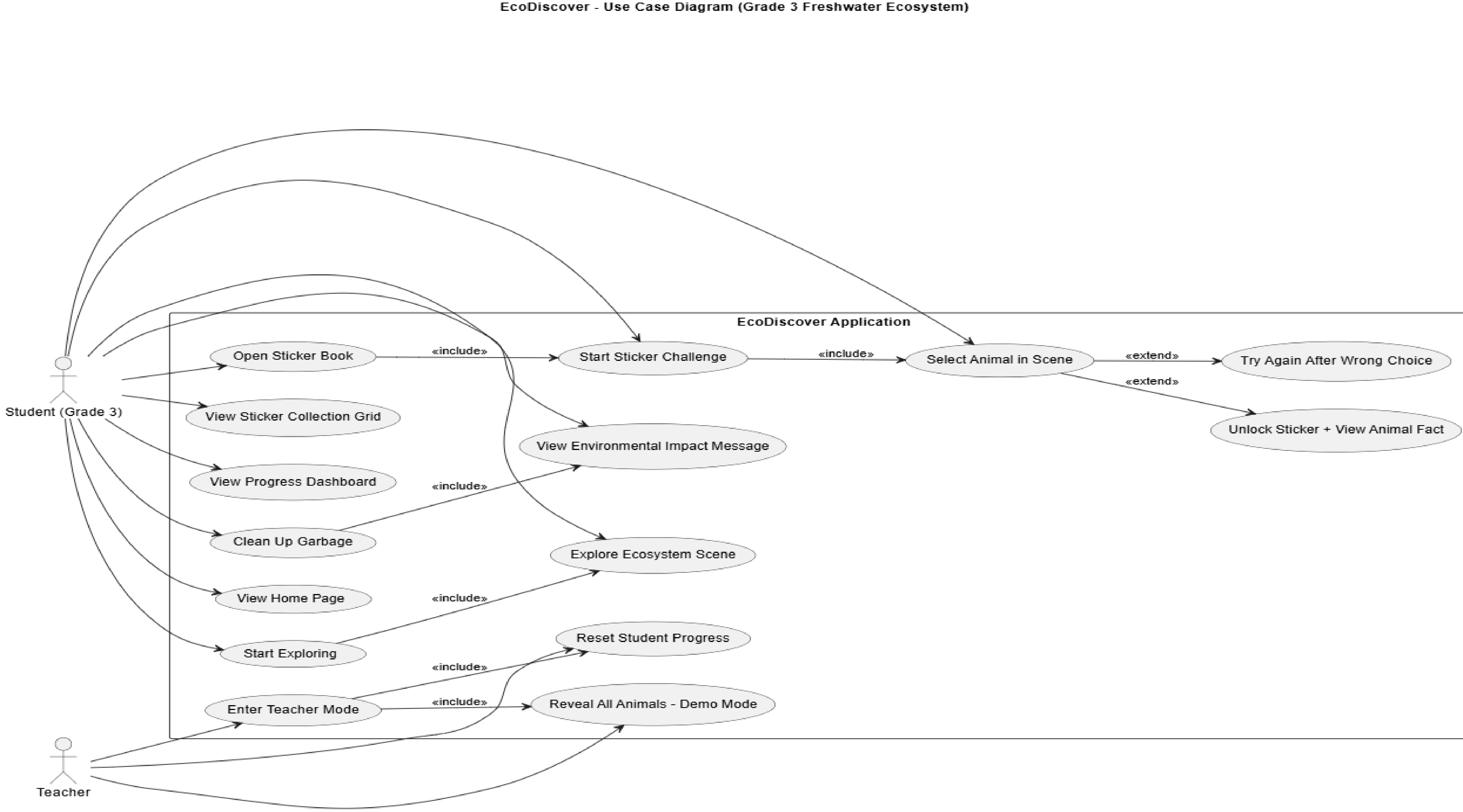
- Handles user actions and updates the system state


MVC Architecture - Ecosystem Discovery Application
Student
View (UI Layer)
Progress Dashboard
Sticker Book
Ecosystem Interface
Home Page
Controller (Logic Layer)
Teacher Controller
Ecosystem Controller
App Controller
Model (Data Layer)
Sticker Book Data
Progress Tracker
Discovery State
Animal Data

UML User Case Diagram

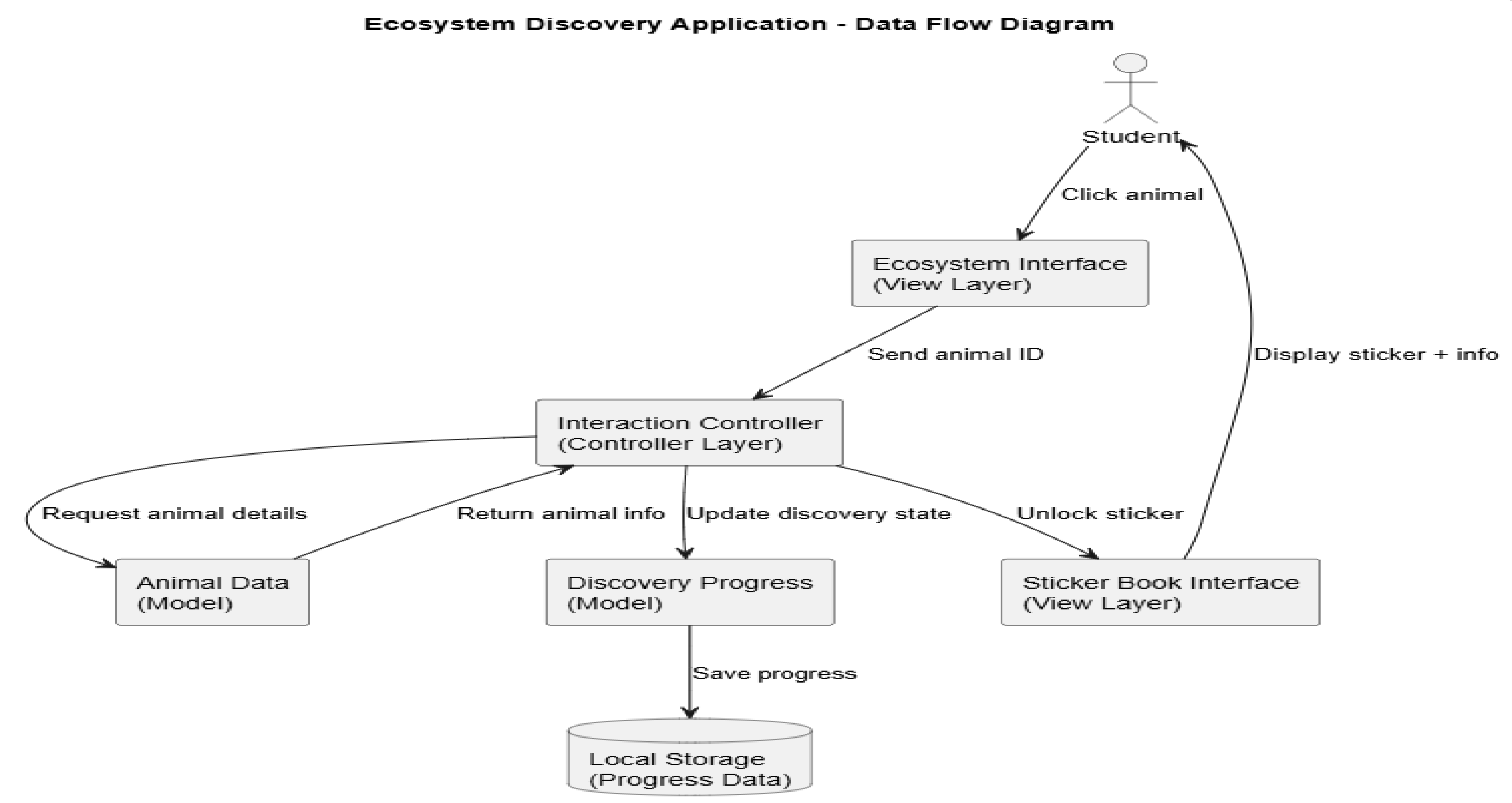
The Use Case Diagram illustrates how different users interact with the EcoDiscovery system and the main functionalities provided by the application.

- **Primary Actors**
- **Student (Primary User)**
  - Explores the ecosystem environment
  - Identifies animals within the ecosystem
  - Unlocks stickers after discovering animals
  - Views progress in the sticker book
- **Teacher (Secondary User)**
  - Accesses teacher mode
  - Resets student progress
  - Reveals all animals for demonstration purposes


EcoDiscover - Use Case Diagram (Grade 3 Freshwater Ecosystem)
EcoDiscover Application
Student (Grade 3)
Teacher
Open Sticker Book
«include»
Start Sticker Challenge
«include»
Select Animal in Scene
«extend»
Try Again After Wrong Choice
«extend»
Unlock Sticker + View Animal Fact
View Sticker Collection Grid
View Environmental Impact Message
View Progress Dashboard
«include»
Clean Up Garbage
Explore Ecosystem Scene
View Home Page
«include»
Start Exploring
Reset Student Progress
«include»
Enter Teacher Mode
«include»
Reveal All Animals - Demo Mode

Data Flow Diagram

- This diagram shows how information moves through the system when a user interacts with the ecosystem.
- Example flow:
- User Interaction
  ↓
  Controller Processes Input
  ↓
  Animal Data Retrieved
  ↓
  Discovery Progress Updated
  ↓
  Sticker Book and Interface Updated


Ecosystem Discovery Application - Data Flow Diagram
Student
Click animal
Ecosystem Interface
(View Layer)
Send animal ID
Display sticker + info
Interaction Controller
(Controller Layer)
Request animal details
Return animal info
Update discovery state
Unlock sticker
Animal Data
(Model)
Discovery Progress
(Model)
Sticker Book Interface
(View Layer)
Save progress
Local Storage
(Progress Data)

High Fidelity Prototype

Get Ready For User Experience through Figma !!

Team Reflections

- This activity helped us understand how software engineering planning and system design are applied within a real-world project context.
- We enjoyed designing an interactive educational concept and developing the high-fidelity prototype.
- One challenge was ensuring consistency across documentation, system diagrams, and the interface design.
- We are most proud of creating a clear system architecture along with a prototype that effectively demonstrates the intended user experience.
- As a team, one of the most difficult tasks was making decisions when different opinions arose and finding a common ground.
- In the future, we plan to continue applying Agile workflows, UML design techniques, and user-centered design approaches.
- We would appreciate further guidance on how to transition from system design into full system implementation.

Feedback

Thank You